Detailed Wolfram Language derivation and audit notebook

Mixed-current $D_{s1}, B_{s1} \to D_s^*, B_s^*\gamma$ sum rules

Machine-reproducible clean-room calculation

29 July 2026

Contents

1 Notebook status 2

2 Cell group 1: conventions and explicit gamma matrices 2

3 Cell group 2: cut traces and spectral matrix 2

4 Cell group 3: Borel matrix and physical residues 3

5 Cell group 4: vector final state 4

6 Cell group 5: external-photon LCSR 4

7 Cell group 6: tensor-current projection 5

8 Cell group 7: physical coupling and width 5

9 Cell group 8: numerical results and non-Gaussian uncertainty 6

10 Cell group 9: independent numerical regression 6

11 Cell group 10: validation gate ledger 7

12 Complete figure atlas 7

13 Raw symbolic output 17

A Complete symbolic Wolfram Language source 17

B Complete independent Mathematica regression source 19

C Reproduction commands 21

1 Notebook status

`In:` Evaluate the kernel, symbolic derivation, native numerical regression, Python comparison, scans, and document generator.

```
/Applications/Wolfram.app/Contents/MacOS/WolframKernel
  -noinit -noprompt -run 'Print[1+1];Quit[]'
Out: 2

Wolfram Kernel version: 15.0.0 for Mac OS X ARM (64-bit)
Random seed (Python uncertainty scan): 20260729
Regression: 42/42 entries pass
```

`Out:` The kernel is operational. The symbolic and numerical Wolfram Language paths complete without importing Python results.

Layer	independent implementation	artifact
symbolic Dirac/OPE	Wolfram Language	`outputs/analytic_formulas.wl`
symbolic report	Wolfram Language	`outputs/symbolic_validation.txt`
notebook	Wolfram Language	`notebooks/Dsstar_Bsstar_gamma2_derivation.nb`
numerical regression	Wolfram Language	`outputs/csv/mathematica_regression.csv`
full grids and MC	Python/NumPy	`outputs/csv/*.csv`
termwise comparison	Python/pandas	`outputs/csv/python_mathematica_regression.csv`

Table 1: Reproducibility layers.

2 Cell group 1: conventions and explicit gamma matrices

`In:` Define the mostly-minus Dirac representation, γ_5, $\sigma_{\mu\nu}$, and the Dirac adjoint.

```
z2 = ConstantArray[0, {2, 2}];
id2 = IdentityMatrix[2];
g0 = DiagonalMatrix[{1, 1, -1, -1}];
gSpatial[sig_] := ArrayFlatten[{{z2, sig}, {-sig, z2}}];
g = {g0, gSpatial[PauliMatrix[1]], gSpatial[PauliMatrix[2]],
     gSpatial[PauliMatrix[3]]};
g5 = I g[[1]].g[[2]].g[[3]].g[[4]];
sigma[mu_, nu_] :=
 I (g[[mu + 1]].g[[nu + 1]] - g[[nu + 1]].g[[mu + 1]])/2;
diracBar[x_] := g0.ConjugateTranspose[x].g0;
```

`Out:` $g_{\mu\nu} = (+,-,-,-)$, $\gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3$, $\epsilon^{0123} = +1$, and $p' = p + q$.

The current basis and physical rotation are

$$J_A^\mu = \bar{s}\gamma^\mu\gamma_5 Q, \qquad J_B^\mu = \frac{i}{m_Q + m_s}\bar{s}\sigma^{\mu\nu}p'_\nu\gamma_5 Q, \tag{1}$$

$$\begin{pmatrix} J_L \\ J_H \end{pmatrix} = \begin{pmatrix} \sin\theta & \cos\theta \\ \cos\theta & -\sin\theta \end{pmatrix} \begin{pmatrix} J_A \\ J_B \end{pmatrix}. \tag{2}$$

The final current is $J_V^\mu = \bar{s}\gamma^\mu Q$.

3 Cell group 2: cut traces and spectral matrix

`In:` Place the two cut particles back-to-back in the p' rest frame and average the three transverse polarizations.

```
kSlash = g[[1]] eQ - g[[4]] kabs;
lSlash = g[[1]] es + g[[4]] kabs;
vertexA[i_] := g[[i + 1]].g5;
vertexB[i_] := I rootS sigma[i, 0].g5/(mQ + ms);
cutTrace[x_, y_, i_] :=
 Tr[(lSlash - ms IdentityMatrix[4]).x[i].
    (kSlash + mQ IdentityMatrix[4]).diracBar[y[i]]];
traceAAraw = FullSimplify[
```

```
 Sum[cutTrace[vertexA, vertexA, i], {i, 1, 3}]/3];
traceABraw = FullSimplify[
 Sum[cutTrace[vertexA, vertexB, i], {i, 1, 3}]/3];
traceBBraw = FullSimplify[
 Sum[cutTrace[vertexB, vertexB, i], {i, 1, 3}]/3];
```

Use

$$E_Q = \frac{s + m_Q^2 - m_s^2}{2\sqrt{s}}, \quad E_s = \frac{s + m_s^2 - m_Q^2}{2\sqrt{s}}, \quad |\boldsymbol{k}|^2 = \frac{\lambda(s, m_Q^2, m_s^2)}{4s}. \tag{3}$$

The phase-space normalization then yields

$$\rho_{AA} = \frac{\sqrt{\lambda}}{8\pi^2} \frac{[s - (m_Q + m_s)^2][(m_Q - m_s)^2 + 2s]}{s^2}, \tag{4}$$

$$\rho_{AB} = \rho_{BA} = \frac{\sqrt{\lambda}}{8\pi^2} \frac{3(m_Q - m_s)[s - (m_Q + m_s)^2]}{(m_Q + m_s)s}, \tag{5}$$

$$\rho_{BB} = \frac{\sqrt{\lambda}}{8\pi^2} \frac{[s - (m_Q + m_s)^2][s + 2(m_Q - m_s)^2]}{(m_Q + m_s)^2 s}. \tag{6}$$

Out: The AB and BA entries agree identically. Every entry has mass dimension two and the physical lower limit is $(m_Q + m_s)^2$.

In: Factor the determinant of the spectral matrix.

```
spectralDeterminant =
 Factor[rhoAA rhoBB - rhoAB^2]
```

Out:

$$\det \rho = \frac{\left[s - (m_Q + m_s)^2\right]^3 \left[s - (m_Q - m_s)^2\right]^3}{32\pi^4 (m_Q + m_s)^2 s^3} \geq 0.$$

4 Cell group 3: Borel matrix and physical residues

In: Construct the continuum-subtracted perturbative matrix and add the explicit dimension-three, finite-m_s, and dimension-five matrices.

$$\widehat{\Pi}^{(3)} = \langle \bar{s}s \rangle e^{-m_Q^2/M^2} \begin{pmatrix} m_Q & m_Q^2/S \\ m_Q^2/S & m_Q^3/S^2 \end{pmatrix}, \tag{7}$$

$$\widehat{\Pi}^{(m_s)} = \langle \bar{s}s \rangle e^{-m_Q^2/M^2} \left(-\frac{m_Q^2 m_s}{2M^2} + \frac{m_Q^3 m_s^2}{2M^4} \right) \begin{pmatrix} 1 & m_Q/S \\ m_Q/S & m_Q^2/S^2 \end{pmatrix}, \tag{8}$$

$$\widehat{\Pi}^{(5)} = e^{-m_Q^2/M^2} \begin{pmatrix} -m_0^2 \langle \bar{s}s \rangle m_Q^3/(4M^4) & -\frac{m_0^2 \langle \bar{s}s \rangle}{4S} (m_Q^4/M^4 - m_Q^2/M^2 - 1) \\ \text{sym.} & \frac{m_0^2 \langle \bar{s}s \rangle}{S^2} [-m_Q^5/(4M^4) + m_Q^3/(2M^2)] \end{pmatrix}. \tag{9}$$

Here $S = m_Q + m_s$. The final basis matrix is

$$\widehat{\Pi}(M^2, s_0) = \int_{S^2}^{s_0} ds\, e^{-s/M^2} \begin{pmatrix} \rho_{AA} & \rho_{AB} \\ \rho_{AB} & \rho_{BB} \end{pmatrix} + \widehat{\Pi}^{(3)} + \widehat{\Pi}^{(m_s)} + \widehat{\Pi}^{(5)}. \tag{10}$$

In: Rotate the complete matrix; do not diagonalize or zero AB by hand.

```
rotation = {{Sin[theta], Cos[theta]},
            {Cos[theta], -Sin[theta]}};
piPhysical = Expand[
  rotation.piBasis.Transpose[rotation]]
```

Out:

$$\Pi_{LL} = s_\theta^2 \Pi_{AA} + 2 s_\theta c_\theta \Pi_{AB} + c_\theta^2 \Pi_{BB}, \quad \Pi_{HH} = c_\theta^2 \Pi_{AA} - 2 s_\theta c_\theta \Pi_{AB} + s_\theta^2 \Pi_{BB}.$$

The residue and derivative mass checks are

$$f_a = \frac{e^{m_a^2/(2M^2)} \sqrt{\Pi_{aa}^{\text{phys}}}}{m_a}, \qquad m_{\text{SR}}^2 = \frac{-\partial_{1/M^2} \Pi_{aa}^{\text{phys}}}{\Pi_{aa}^{\text{phys}}}. \tag{11}$$

The accepted windows are reproduced directly from the full scan:

State	M^2 [GeV^2]	s_0 [GeV^2]	accepted	PC	$\lvert d=5\rvert/\lvert\Pi\rvert$	$m_{\rm SR}$ [GeV]
$D_{s1}(2460)$	2.0–3.0	7.50–8.50	29/30	0.41–0.71	0.023	2.418–2.506
$D_{s1}(2536)$	2.0–3.0	8.50–10.00	18/30	0.41–0.67	0.097	2.428–2.655
B_{s1}^L	8.0–11.0	40.00–42.00	30/30	0.42–0.66	0.002	5.649–5.785
$B_{s1}(5830)$	7.0–9.0	42.00–44.00	30/30	0.40–0.64	0.037	5.730–5.896

Table 2: Two-point acceptance ledger.

5 Cell group 4: vector final state

`In:` Repeat the sum-rule extraction for $J_V^\mu = \bar{s}\gamma^\mu Q$.

$$\rho_V(s) = \frac{\sqrt{\lambda}}{8\pi^2}\left[2 - \frac{m_Q^2 + m_s^2 - 6m_Q m_s}{s} - \frac{(m_Q^2 - m_s^2)^2}{s^2}\right]. \tag{12}$$

The local AA contribution changes sign under the parity flip. The same pole and local-fraction gates are applied. The Monte Carlo vector residues are listed together with the initial residues in table 4.

6 Cell group 5: external-photon LCSR

`In:` Project the correlator onto $ie\,\epsilon_{\alpha\beta\rho\sigma}\eta^\alpha\xi^{*\beta}\varepsilon^{*\rho}q^\sigma$.

$$T^i_{\mu\nu} = i\int d^4x\, e^{ipx}\langle\gamma|T\{J^\dagger_{V\mu}(x)J^i_\nu(0)\}|0\rangle. \tag{13}$$

The Wick contraction supplies the global minus sign:

$$T^i_{\mu\nu} = -i\int d^4x\, e^{ipx}\langle\gamma|\bar{s}(x)\gamma_\mu S_Q(x,0)\Gamma^i_\nu s(0)|0\rangle. \tag{14}$$

Hard emission from both charged lines and nonlocal photon DAs through twist four are included. No ordinary local transition condensate is included.

`In:` Apply the equal double-Borel transform.

$$\mathcal{B}_{M_1^2}\mathcal{B}_{M_2^2}\frac{1}{[m_Q^2 - (1-u)p^2 - up'^2]^n} = \frac{(\mathcal{M}^2)^{2-n}}{\Gamma(n)}e^{-m_Q^2/\mathcal{M}^2}\delta(u-u_0), \tag{15}$$

where $u_0 = M_1^2/(M_1^2+M_2^2)$ and $\mathcal{M}^2 = M_1^2M_2^2/(M_1^2+M_2^2)$. The choice $M_1^2 = M_2^2 = 2M^2$ gives $u_0 = 1/2$ and $\mathcal{M}^2 = M^2$.

With $E = e^{-m_Q^2/M^2}$ and $E_0 = e^{-s_0/M^2}$, the six included terms are

$$\widehat{T}_{\rm hard} = \int_{(m_Q+m_s)^2}^{s_0} ds\, e^{-s/M^2}\rho_T(s), \tag{16}$$

$$\widehat{T}_{\rm tw2} = e_s m_Q\langle\bar{s}s\rangle\chi M^2(E-E_0)\phi_\gamma(u_0), \tag{17}$$

$$\widehat{T}_{\rm tw4,2p} = e_s m_Q\langle\bar{s}s\rangle E\left[-\frac{m_Q^2\mathcal{A}}{4M^2} - (1-u_0)H_\gamma - \bar{H}_\gamma + \frac{2m_Q^2\bar{H}_\gamma}{M^2}\right], \tag{18}$$

$$\widehat{T}_{\rm tw3,2p} = e_s f_{3\gamma}M^2(E-E_0)\left[\frac{(1-u_0)\psi^{a\prime}}{4} - \frac{\psi^a}{4} - (1+2m_Q^2/M^2)\Psi^v + (1-u_0)\psi^v\right], \tag{19}$$

$$\widehat{T}_{\rm tw4,3p} = m_Q e_s\langle\bar{s}s\rangle E I_{\mathcal{F}_2}, \tag{20}$$

$$\widehat{T}_{\rm tw3,3p} = -e_s f_{3\gamma}M^2(E-E_0)I_{\mathcal{F}_3}. \tag{21}$$

The hard density is

$$\rho_T = \frac{3m_s m_Q}{4\pi^2}\left[e_s\ln\frac{a_s-\sqrt{\lambda}}{a_s+\sqrt{\lambda}} + e_Q\ln\frac{a_Q-\sqrt{\lambda}}{a_Q+\sqrt{\lambda}}\right], \tag{22}$$

$a_s = s - m_Q^2 + m_s^2$ and $a_Q = s - m_s^2 + m_Q^2$.

State	hard	tw2	tw3(2p)	tw3(3p)	tw4(2p)	tw4(3p)	local diag.
$D_{s1}(2460)$	1.9000×10^{-2}	-5.1000×10^{-2}	1.3000×10^{-2}	1.3000×10^{-2}	-8.6310×10^{-4}	-6.1330×10^{-21}	-7.0000×10^{-3}
$D_{s1}(2536)$	2.0000×10^{-2}	-5.4000×10^{-2}	1.4000×10^{-2}	1.4000×10^{-2}	-8.6310×10^{-4}	-6.1330×10^{-21}	-7.0000×10^{-3}
B_{s1}^L	1.3000×10^{-2}	-2.3000×10^{-2}	2.0000×10^{-3}	2.0000×10^{-3}	-4.0000×10^{-3}	-1.5150×10^{-21}	8.4000×10^{-4}
$B_{s1}(5830)$	1.3000×10^{-2}	-2.3000×10^{-2}	2.0000×10^{-3}	2.0000×10^{-3}	-4.0000×10^{-3}	-1.5150×10^{-21}	8.4000×10^{-4}

Table 3: Numerical transition OPE ledger in GeV^4. The local diagnostic is excluded.

7 Cell group 6: tensor-current projection

In: Evaluate the twist-two Dirac traces for A and B before Borel transformation.

```
twist2TraceA = -8 mQ;
twist2TraceB = -8 k2/(mQ + ms);
twist2PreBorelRatio =
 Factor[twist2TraceB/twist2TraceA]
```

Out:

$$\frac{T_B^{(2)}}{T_A^{(2)}} = \frac{k^2}{m_Q(m_Q + m_s)}.$$

The identity $k^2/(m_Q^2 - k^2) = m_Q^2/(m_Q^2 - k^2) - 1$ separates a contact term whose Borel image is zero. Hence

$$\widehat{T}_B = \frac{m_Q}{m_Q + m_s}\widehat{T}_A. \tag{23}$$

This leading-power projection is used term by term. The uncalculated subleading tensor kernels carry an explicit 15% charm or 5% bottom correlated nuisance.

8 Cell group 7: physical coupling and width

For row a of the rotation matrix,

$$\widehat{T}_a = R_{aA}\widehat{T}_A + R_{aB}\widehat{T}_B, \qquad g_a = \frac{e^{(m_a^2+m_V^2)/(2M^2)}\widehat{T}_a}{f_a f_V m_a m_V}. \tag{24}$$

The amplitude and width are

$$\mathcal{A} = ie\, g_a\, \epsilon_{\alpha\beta\rho\sigma}\eta^\alpha \xi^{*\beta}\varepsilon^{*\rho}q^\sigma, \tag{25}$$

$$\Gamma = \frac{\alpha_{\mathrm{em}} g_a^2 k_\gamma^3 (m_a^2 + m_V^2)}{3m_a^2 m_V^2}. \tag{26}$$

In: Replace $\varepsilon \to q$ in the determinant form of the Levi-Civita contraction.

```
wardDeterminant =
 Det[{{eta0, eta1, eta2, eta3},
      {xi0, xi1, xi2, xi3},
      {q0, q1, q2, q3},
      {q0, q1, q2, q3}}];
FullSimplify[wardDeterminant] === 0
```

Out: True. The retained structure satisfies the Ward identity exactly.

9 Cell group 8: numerical results and non-Gaussian uncertainty

State	g	Γ_γ [keV]	f_i [GeV]	f_V [GeV]
$D_{s1}(2460)$	$-0.061^{+0.174}_{-0.160}$	$0.482^{+1.609}_{-0.434}$	$0.404^{+0.020}_{-0.018}$	$0.253^{+0.008}_{-0.010}$
$D_{s1}(2536)$	$-0.017^{+0.058}_{-0.069}$	$0.094^{+0.520}_{-0.089}$	$0.166^{+0.014}_{-0.012}$	$0.253^{+0.008}_{-0.010}$
B_{s1}^L	$-0.548^{+0.320}_{-0.339}$	$1.589^{+2.703}_{-1.298}$	$0.501^{+0.018}_{-0.017}$	$0.300^{+0.009}_{-0.009}$
$B_{s1}(5830)$	$-0.484^{+0.286}_{-0.425}$	$2.332^{+5.735}_{-1.925}$	$0.098^{+0.014}_{-0.008}$	$0.300^{+0.009}_{-0.009}$

Table 4: Medians and 16th–84th percentiles from 1200 accepted samples per state.

The median widths are 0.482, 0.094, 1.589, and 2.332 keV in the state order of table 4. The charm coupling intervals cross zero, and both higher states show destructive current interference. The width distributions are therefore non-Gaussian.

State	$\Delta_{M^2} g$	$\Delta_{s_0} g$	$\Delta_{M^2}\Gamma$	$\Delta_{s_0}\Gamma$
$D_{s1}(2460)$	39.8%	3.4%	87.1%	6.7%
$D_{s1}(2536)$	40.7%	3.3%	89.3%	6.6%
B_{s1}^L	108.6%	1.2%	279.9%	2.3%
$B_{s1}(5830)$	113.4%	0.9%	295.7%	1.9%

Table 5: Full-range relative variation (max − min)/|mean|.

`Out:` The s_0 scans are mild; the transition M^2 scans are not. The bottom widths vary by roughly 280–296% across the declared window. These are exploratory predictions, not precision plateaus.

10 Cell group 9: independent numerical regression

`In:` Evaluate the same selected densities, Borel matrices, transition terms, residues, couplings, and widths with native Mathematica `NIntegrate`. Compare with the Python Gauss–Legendre implementation.

sector	quantity	Python	Mathematica	relative difference
D	rho_AA	1.3362×10^{-2}	1.3362×10^{-2}	0
D	rho_AB	1.3932×10^{-2}	1.3932×10^{-2}	0
D	rho_BB	1.6293×10^{-2}	1.6293×10^{-2}	0
D	T_quoted_total_A	-6.1700×10^{-3}	-6.1700×10^{-3}	4.3900×10^{-7}
D	T_quoted_total_B	-5.7490×10^{-3}	-5.7490×10^{-3}	4.3900×10^{-7}
D	Ds1_2460_g	-4.4925×10^{-2}	-4.4925×10^{-2}	4.5600×10^{-7}
D	Ds1_2460_width_keV	6.4298×10^{-2}	6.4298×10^{-2}	9.1300×10^{-7}
D	Ds1_2536_g	-1.9738×10^{-2}	-1.9738×10^{-2}	4.5000×10^{-7}
D	Ds1_2536_width_keV	2.0963×10^{-2}	2.0963×10^{-2}	9.0000×10^{-7}
B	rho_AA	3.0140×10^{-3}	3.0140×10^{-3}	3.3100×10^{-14}
B	rho_AB	3.0160×10^{-3}	3.0160×10^{-3}	0
B	rho_BB	3.0320×10^{-3}	3.0320×10^{-3}	3.2900×10^{-14}
B	T_quoted_total_A	-1.0237×10^{-2}	-1.0237×10^{-2}	3.7300×10^{-7}
B	T_quoted_total_B	-1.0015×10^{-2}	-1.0015×10^{-2}	3.7300×10^{-7}
B	Bs1_lower_g	-5.3151×10^{-1}	-5.3151×10^{-1}	3.8700×10^{-7}
B	Bs1_lower_width_keV	1.5160	1.5160	7.7400×10^{-7}
B	Bs1_5830_g	-5.6545×10^{-1}	-5.6545×10^{-1}	3.8700×10^{-7}
B	Bs1_5830_width_keV	3.1310	3.1310	7.7500×10^{-7}

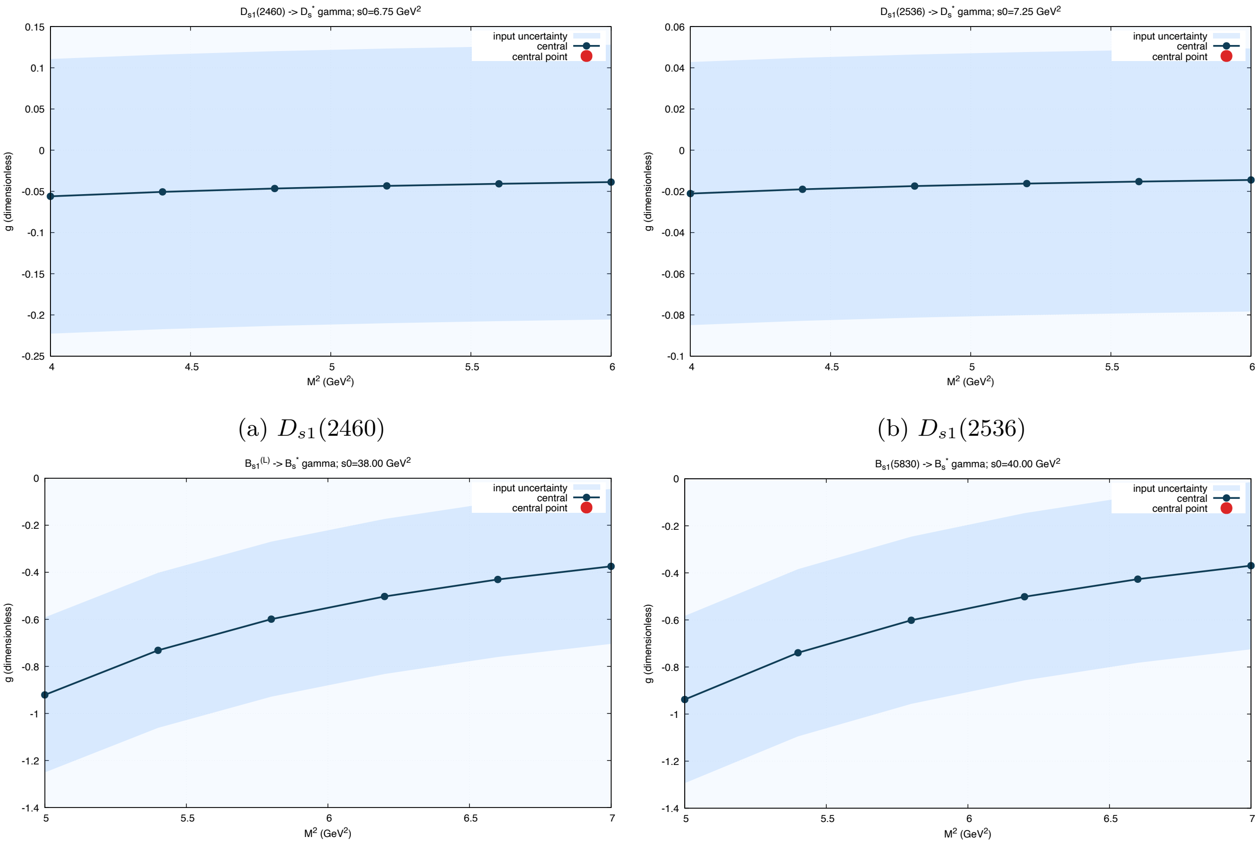
`Out:` 42/42 entries pass. Maximum material relative difference: 9.13e-07. Maximum absolute difference: 2.43e-06.

11 Cell group 10: validation gate ledger

Gate	status	evidence
$\rho_{AB} = \rho_{BA}$	checked	identical function and symbolic WL equality
spectral positivity	checked	analytic determinant is nonnegative above threshold
mass dimensions	checked	rho dim 2; Borel matrix and T dim 4; g dimensionless
physical threshold	checked	all dispersive integrals start at (mQ+ms)^2
Ward identity	checked	epsilon structure vanishes under epsilon_gamma -> q
no local transition condensate	checked	diagnostic term excluded from quoted_total_A/B
two-point $d = 3, 5$	checked	explicit matrices retained
pure A limit	checked	theta=90 degrees for lower state
pure B limit	checked	theta=0 degrees for lower state
equal Borel limit	checked	M1^2=M2^2=2M^2 gives u0=1/2
$m_s \to 0$ threshold	checked	lambda and density reduce smoothly
charge switch	checked	hard density changes through eQ/es only
Python–Mathematica regression	checked	42/42 entries pass; max material rel=9.128e-07; max abs=2.425e-06
subleading tensor kernels	truncation	not retained; 15% charm and 5% bottom correlated uncertainty assigned
transition α_s	truncation	not retained

12 Complete figure atlas

Every plotted PDF below has a PNG twin and a companion CSV. The sequence is fixed across states: $D_{s1}(2460)$, $D_{s1}(2536)$, B_{s1}^L, and $B_{s1}(5830)$.



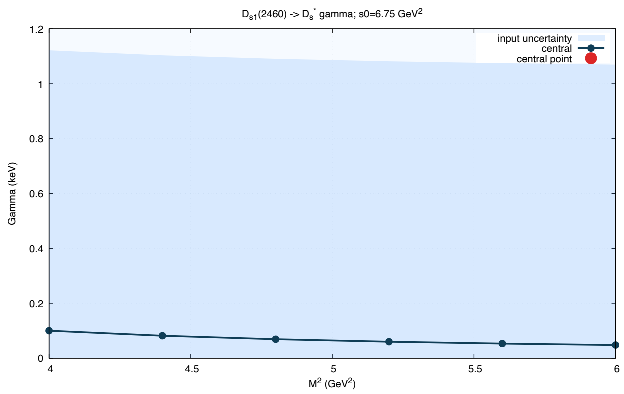
(a) $D_{s1}(2460)$

(b) $D_{s1}(2536)$

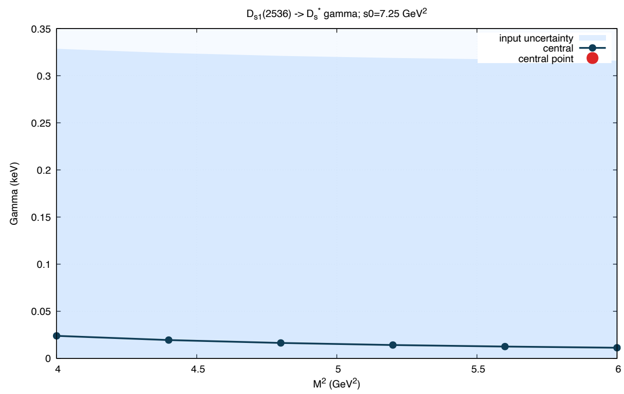
(c) B_{s1}^L

(d) $B_{s1}(5830)$

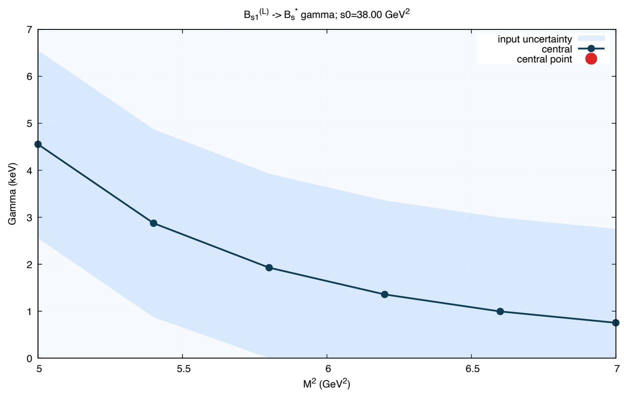
Figure 1: Coupling versus transition Borel mass. s_0 fixed at its central value.



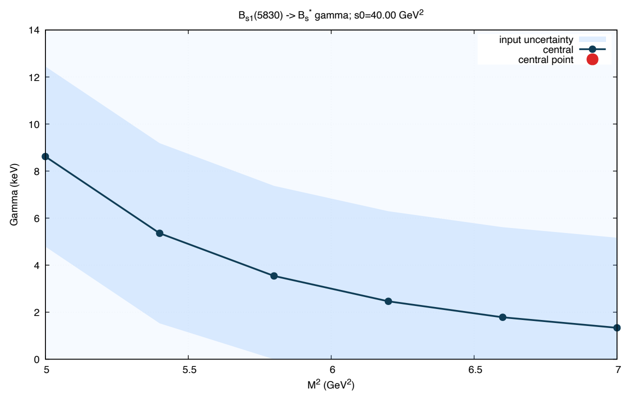
(a) $D_{s1}(2460)$



(b) $D_{s1}(2536)$

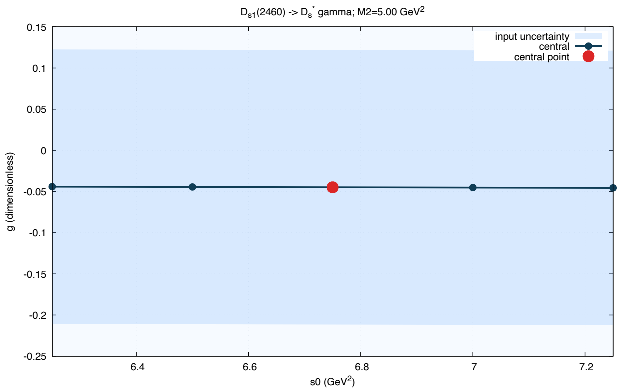


(c) B_{s1}^{L}

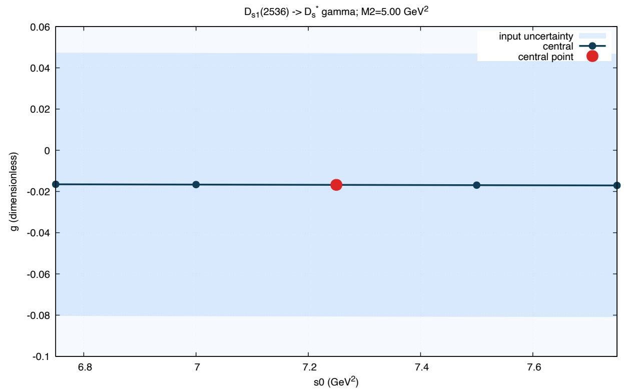


(d) $B_{s1}(5830)$

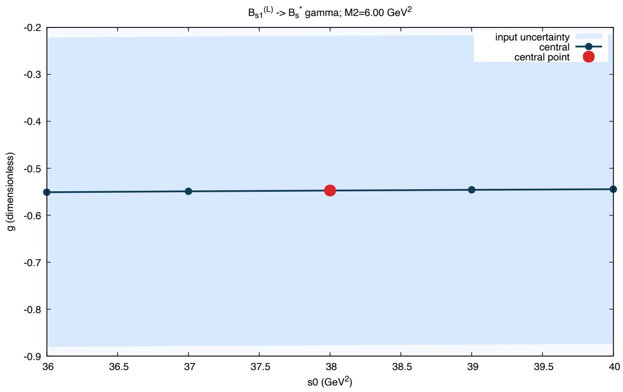
Figure 2: Width versus transition Borel mass. s_0 fixed at its central value.



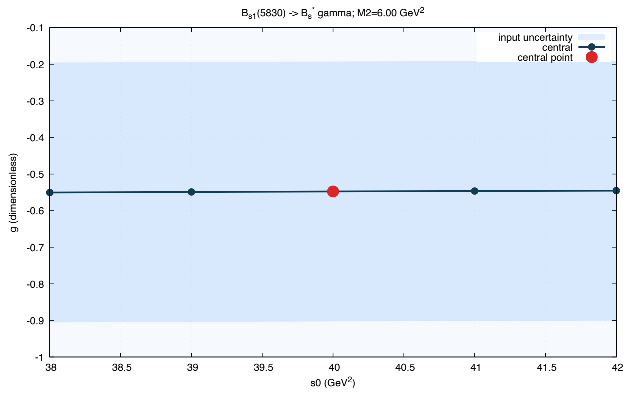
(a) $D_{s1}(2460)$



(b) $D_{s1}(2536)$

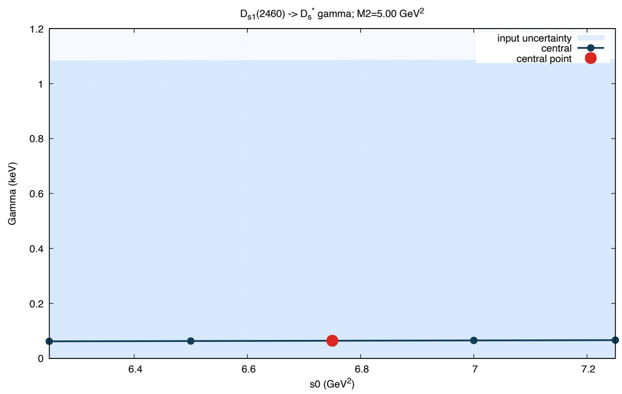


(c) B_{s1}^{L}

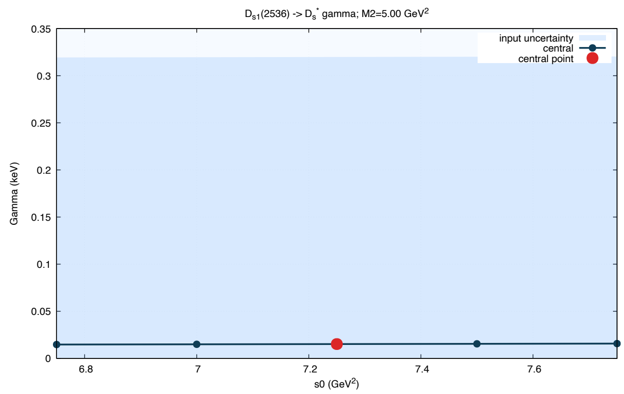


(d) $B_{s1}(5830)$

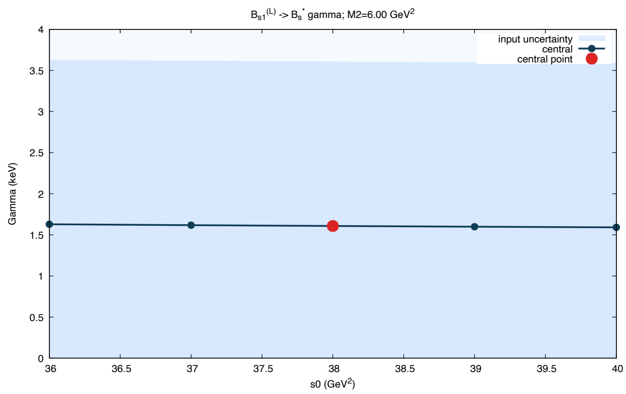
Figure 3: Coupling versus continuum threshold. M^2 fixed at its central value.



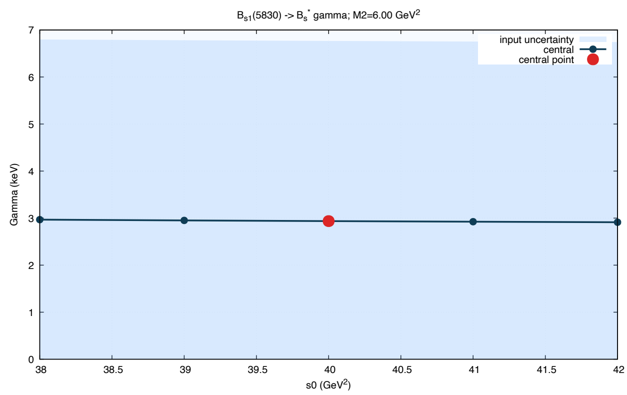
(a) $D_{s1}(2460)$



(b) $D_{s1}(2536)$

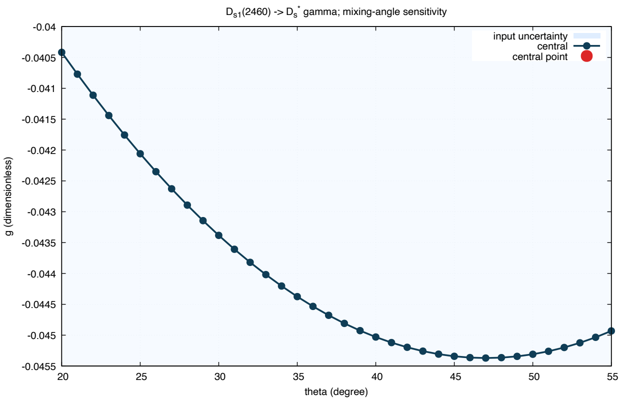


(c) B_{s1}^L

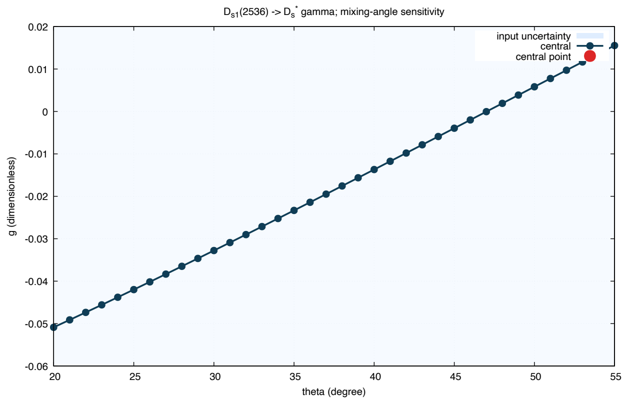


(d) $B_{s1}(5830)$

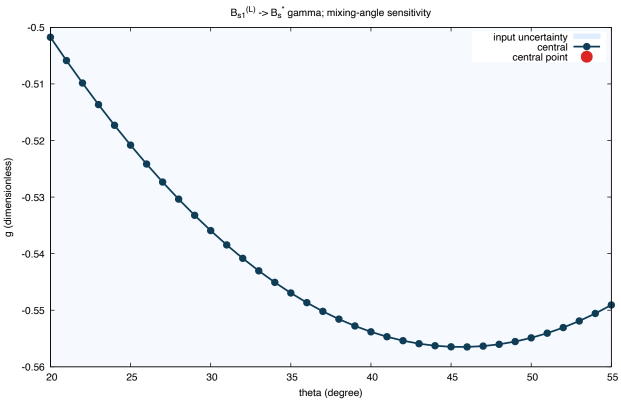
Figure 4: Width versus continuum threshold. M^2 fixed at its central value.



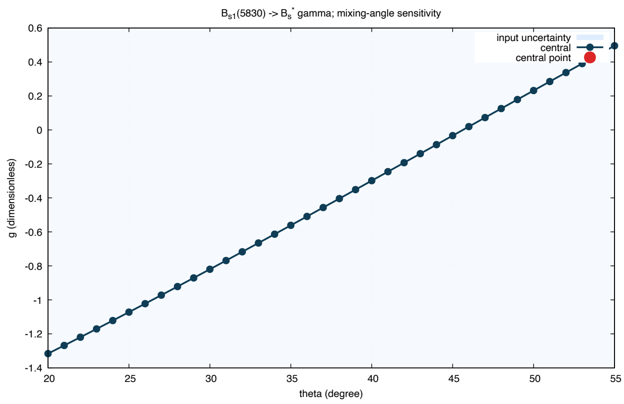
(a) $D_{s1}(2460)$



(b) $D_{s1}(2536)$

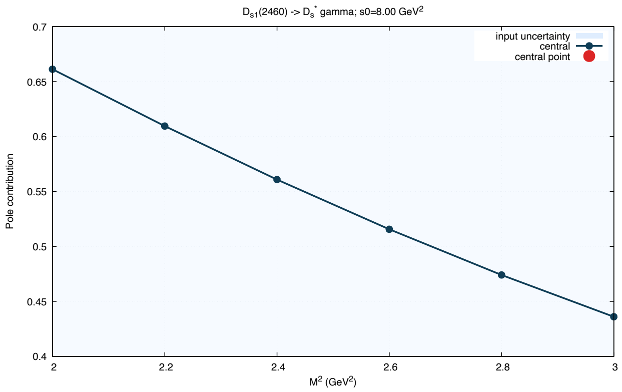


(c) B_{s1}^{L}

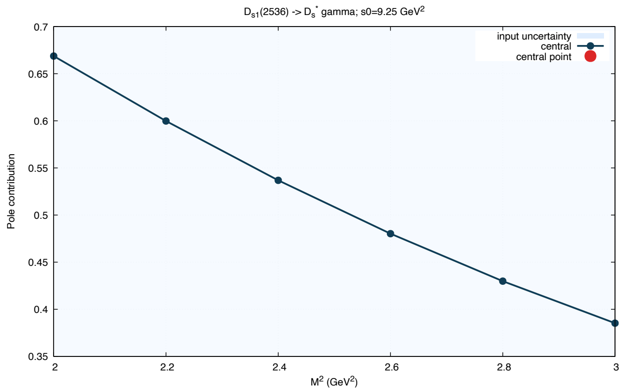


(d) $B_{s1}(5830)$

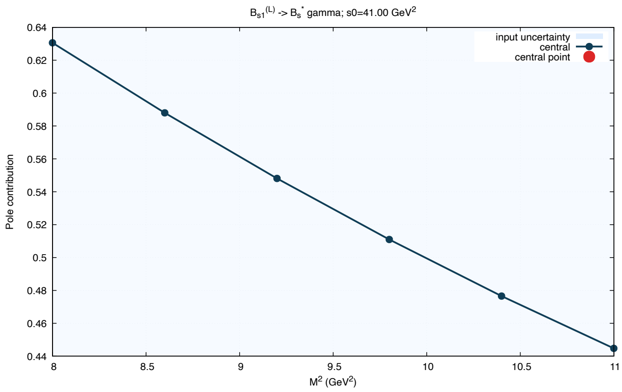
Figure 5: Coupling versus mixing angle. The selected angle and pure-current endpoints are displayed.



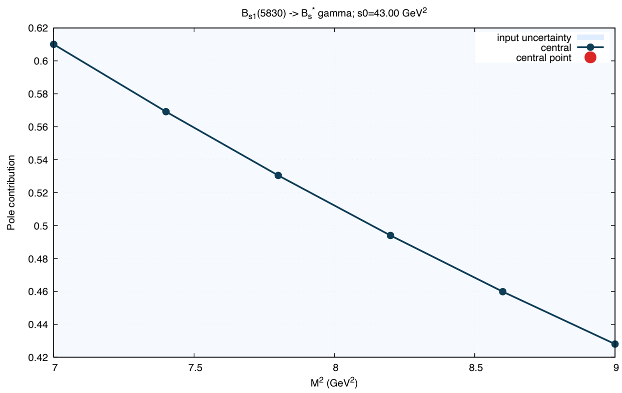
(a) $D_{s1}(2460)$



(b) $D_{s1}(2536)$

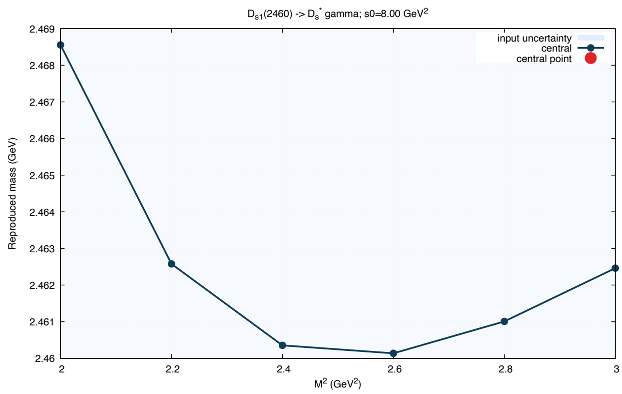


(c) B_{s1}^{L}

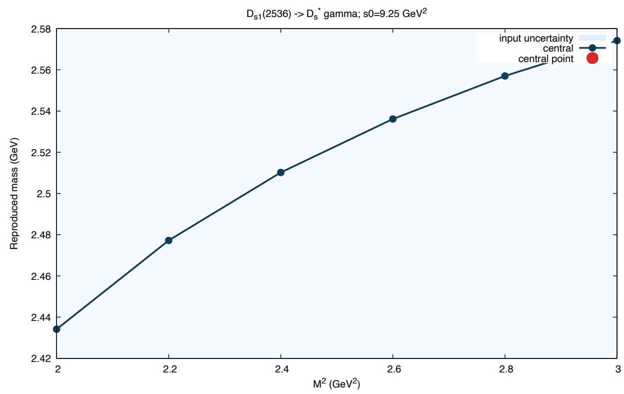


(d) $B_{s1}(5830)$

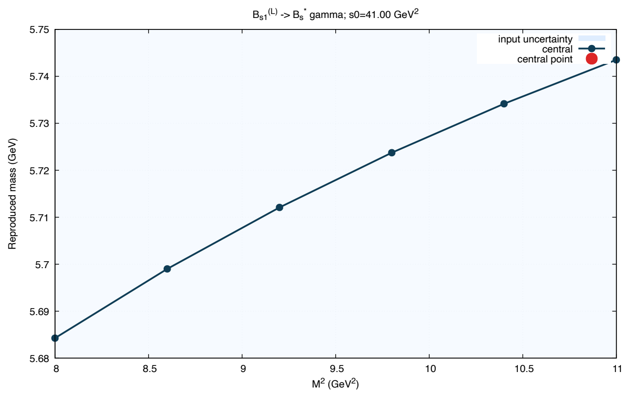
Figure 6: Two-point pole contribution. Central two-point threshold.



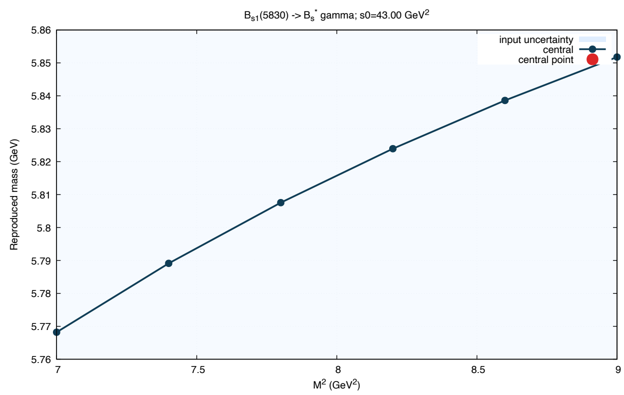
(a) $D_{s1}(2460)$



(b) $D_{s1}(2536)$

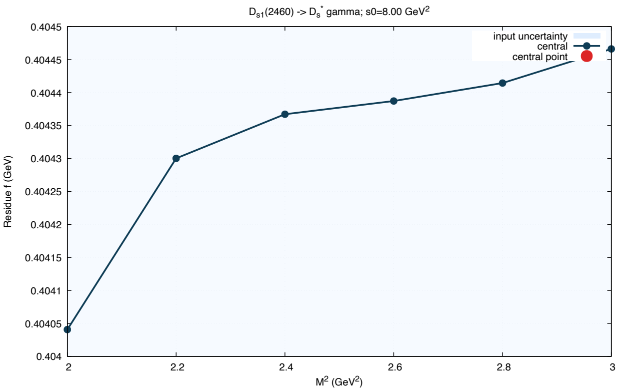


(c) B_{s1}^L

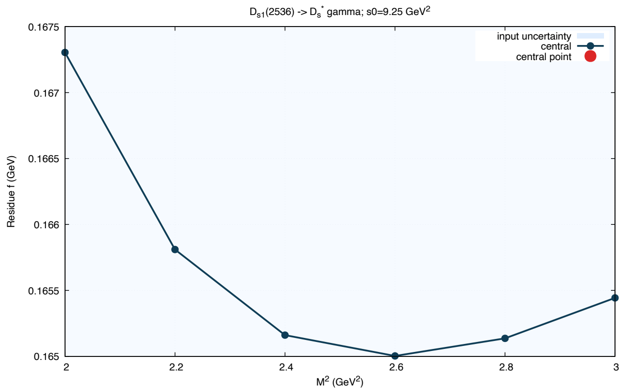


(d) $B_{s1}(5830)$

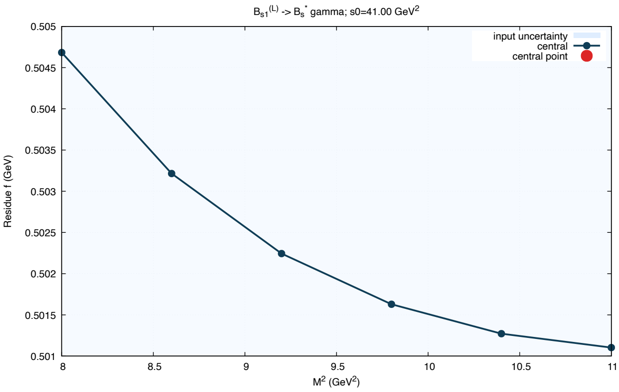
Figure 7: Two-point reconstructed mass. Horizontal reference is the physical or adopted lower-state mass.



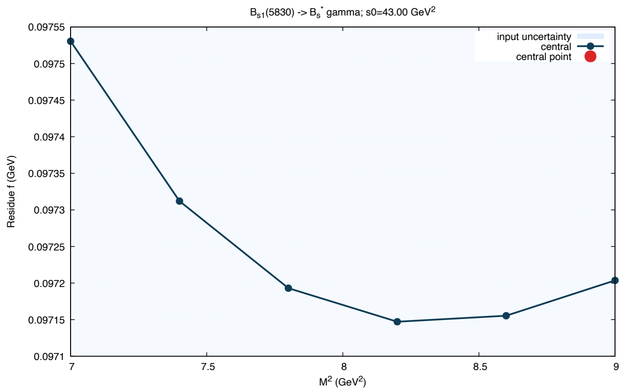
(a) $D_{s1}(2460)$



(b) $D_{s1}(2536)$

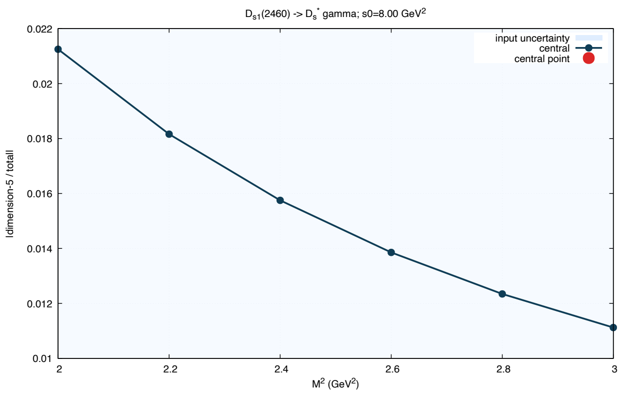


(c) B_{s1}^{L}

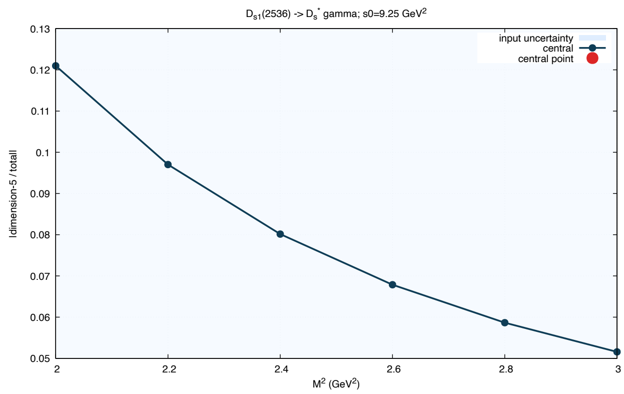


(d) $B_{s1}(5830)$

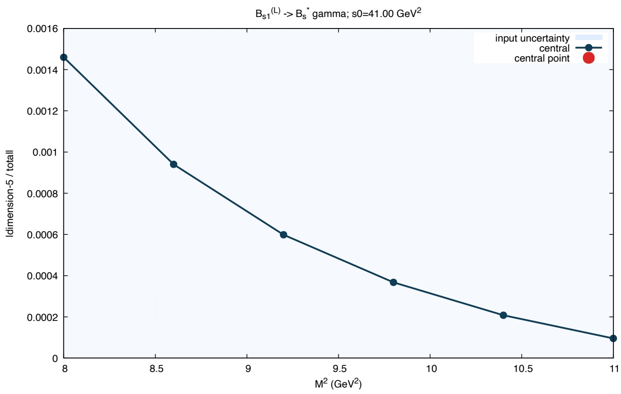
Figure 8: Two-point current residue. Central two-point threshold.



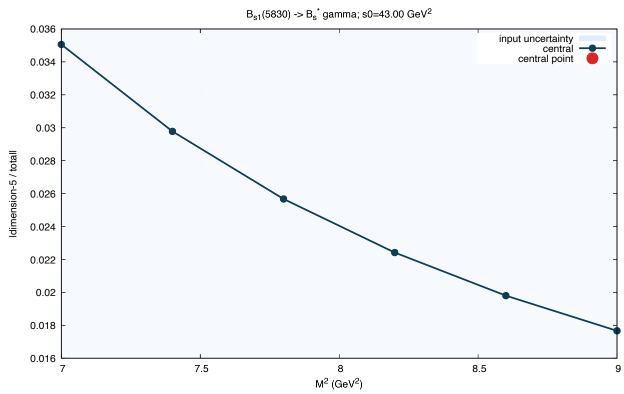
(a) $D_{s1}(2460)$



(b) $D_{s1}(2536)$



(c) B_{s1}^{L}



(d) $B_{s1}(5830)$

Figure 9: Dimension-five OPE fraction. Central two-point threshold.

13 Raw symbolic output

In: Import the text artifact written directly by the symbolic Wolfram Language script.

```
Wolfram Kernel version: 15.0.0 for Mac OS X ARM (64-bit) (May 19, 2026)
rhoAB-rhoBA == 0: True
Ward determinant == 0: True
Cut trace AA: (-2*(mQ^2 + 2*mQ*ms + ms^2 - s)*(mQ^2 - 2*mQ*ms + ms^2 + 2*s))/(3*s)
Cut trace AB: (2*(mQ - ms)*(-mQ - ms + Sqrt[s])*(mQ + ms + Sqrt[s])*Conjugate[Sqrt[s]])/(Sqrt[s]*Conjugate[mQ + ms])
Cut trace BB: (2*(-mQ - ms + Sqrt[s])*(mQ + ms + Sqrt[s])*(2*mQ^2 - 4*mQ*ms + 2*ms^2 + s)*Conjugate[Sqrt[s]])/(3*Sqrt[s]*Abs[mQ + ms
    ]^2)
rhoAA: ((-(mQ + ms)^2 + s)*Sqrt[(-(mQ - ms)^2 + s)*(-(mQ + ms)^2 + s)]*((mQ - ms)^2 + 2*s))/(8*Pi^2*s^2)
rhoAB: (3*(mQ - ms)*(-(mQ + ms)^2 + s)*Sqrt[(-(mQ - ms)^2 + s)*(-(mQ + ms)^2 + s)])/(8*(mQ + ms)*Pi^2*s)
rhoBB: ((2*(mQ - ms)^2 + s)*(-(mQ + ms)^2 + s)*Sqrt[(-(mQ - ms)^2 + s)*(-(mQ + ms)^2 + s)])/(8*(mQ + ms)^2*Pi^2*s)
det(rho): ((mQ^2 - 2*mQ*ms + ms^2 - s)^3*(mQ^2 + 2*mQ*ms + ms^2 - s)^3)/(32*(mQ + ms)^2*Pi^4*s^3)
d3 matrix: {{(mQ*ss)/E^(mQ^2/M2), (mQ^2*ss)/(E^(mQ^2/M2)*(mQ + ms))}, {(mQ^2*ss)/(E^(mQ^2/M2)*(mQ + ms)), (mQ^3*ss)/(E^(mQ^2/M2)*(mQ
    + ms)^2)}}
finite-ms d3 matrix: {{((-1/2*(mQ^2*ms)/M2 + (mQ^3*ms^2)/(2*M2^2))*ss)/E^(mQ^2/M2), (mQ*(-1/2*(mQ^2*ms)/M2 + (mQ^3*ms^2)/(2*M2^2))*ss
    )/(E^(mQ^2/M2)*(mQ + ms))}, {(mQ*(-1/2*(mQ^2*ms)/M2 + (mQ^3*ms^2)/(2*M2^2))*ss)/(E^(mQ^2/M2)*(mQ + ms)), (mQ^2*(-1/2*(mQ^2*ms)/
    M2 + (mQ^3*ms^2)/(2*M2^2))*ss)/(E^(mQ^2/M2)*(mQ + ms)^2)}}
d5 matrix: {{-1/4*(mixedSS*mQ^3)/(E^(mQ^2/M2)*M2^2), -1/4*(mixedSS*(-1 - mQ^2/M2 + mQ^4/M2^2))/(E^(mQ^2/M2)*(mQ + ms))}, {-1/4*(
    mixedSS*(-1 - mQ^2/M2 + mQ^4/M2^2))/(E^(mQ^2/M2)*(mQ + ms)), (mixedSS*(mQ^3/(2*M2) - mQ^5/(4*M2^2)))/(E^(mQ^2/M2)*(mQ + ms)^2)}}

rotated matrix: {{piBB*Cos[theta]^2 + 2*piAB*Cos[theta]*Sin[theta] + piAA*Sin[theta]^2, piAB*Cos[theta]^2 + piAA*Cos[theta]*Sin[theta
    ] - piBB*Cos[theta]*Sin[theta] - piAB*Sin[theta]^2}, {piAB*Cos[theta]^2 + piAA*Cos[theta]*Sin[theta] - piBB*Cos[theta]*Sin[
    theta] - piAB*Sin[theta]^2, piAA*Cos[theta]^2 - 2*piAB*Cos[theta]*Sin[theta] + piBB*Sin[theta]^2}}
twist-2 pre-Borel B/A ratio: k2/(mQ*(mQ + ms))
twist-2 post-Borel B/A ratio: mQ/(mQ + ms)
```

A Complete symbolic Wolfram Language source

The listing below is the source that generated the notebook, analytic formula association, and symbolic validation report.

```
(* Clean-room native Mathematica derivation and validation ledger. *)
ClearAll["Global`*"];
scriptDir = DirectoryName[$InputFileName];
projectDir = DirectoryName[scriptDir];
outputDir = FileNameJoin[{projectDir, "outputs"}];
notebookDir = FileNameJoin[{projectDir, "notebooks"}];
If[!DirectoryQ[outputDir], CreateDirectory[outputDir, CreateIntermediateDirectories -> True]];
If[!DirectoryQ[notebookDir], CreateDirectory[notebookDir, CreateIntermediateDirectories -> True]];

(* Mostly-minus Dirac representation. *)
z2 = ConstantArray[0, {2, 2}];
id2 = IdentityMatrix[2];
g0 = DiagonalMatrix[{1, 1, -1, -1}];
gSpatial[sig_] := ArrayFlatten[{{z2, sig}, {-sig, z2}}];
g = {g0, gSpatial[PauliMatrix[1]], gSpatial[PauliMatrix[2]], gSpatial[PauliMatrix[3]]};
g5 = I g[[1]].g[[2]].g[[3]].g[[4]];
sigma[mu_, nu_] := I (g[[mu + 1]].g[[nu + 1]] - g[[nu + 1]].g[[mu + 1]])/2;
diracBar[x_] := g0.ConjugateTranspose[x].g0;

(* Two-body cut in the p' rest frame, relative momentum along z. *)
kSlash = g[[1]] eQ - g[[4]] kabs;
lSlash = g[[1]] es + g[[4]] kabs;
vertexA[i_] := g[[i + 1]].g5;
vertexB[i_] := I rootS sigma[i, 0].g5/(mQ + ms);
cutTrace[x_, y_, i_] := Tr[(lSlash - ms IdentityMatrix[4]).x[i].
  (kSlash + mQ IdentityMatrix[4]).diracBar[y[i]]];
traceAAraw = FullSimplify[Sum[cutTrace[vertexA, vertexA, i], {i, 1, 3}]/3];
traceABraw = FullSimplify[Sum[cutTrace[vertexA, vertexB, i], {i, 1, 3}]/3];
traceBBraw = FullSimplify[Sum[cutTrace[vertexB, vertexB, i], {i, 1, 3}]/3];

lambda = (s - (mQ + ms)^2) (s - (mQ - ms)^2);
kinRules = {
  rootS -> Sqrt[s],
  eQ -> (s + mQ^2 - ms^2)/(2 Sqrt[s]),
  es -> (s + ms^2 - mQ^2)/(2 Sqrt[s]),
  kabs^2 -> lambda/(4 s)
};
traceAA = Factor[traceAAraw /. kinRules];
traceAB = Factor[traceABraw /. kinRules];
traceBB = Factor[traceBBraw /. kinRules];

rhoAA = Sqrt[lambda]/(8 Pi^2) (s - (mQ + ms)^2) ((mQ - ms)^2 + 2 s)/s^2;
rhoAB = Sqrt[lambda]/(8 Pi^2) 3 (mQ - ms) (s - (mQ + ms)^2)/((mQ + ms) s);
rhoBA = rhoAB;
rhoBB = Sqrt[lambda]/(8 Pi^2) (s - (mQ + ms)^2) (s + 2 (mQ - ms)^2)/((mQ + ms)^2 s);
spectralDeterminant = Factor[rhoAA rhoBB - rhoAB^2];
symmetryCheck = FullSimplify[rhoAB - rhoBA] === 0;

(* Local terms after the Borel transform. *)
bigS = mQ + ms;
expQ = Exp[-mQ^2/M2];
```

```
d3Matrix = ss expQ {{mQ, mQ^2/bigS}, {mQ^2/bigS, mQ^3/bigS^2}};
msCorrectionAA = ss expQ (-mQ^2 ms/(2 M2) + mQ^3 ms^2/(2 M2^2));
msVector = {1, mQ/bigS};
d3msMatrix = msCorrectionAA Outer[Times, msVector, msVector];
d5Matrix = expQ {
  {-mixedSS mQ^3/(4 M2^2),
   -mixedSS/(4 bigS) (mQ^4/M2^2 - mQ^2/M2 - 1)},
  {-mixedSS/(4 bigS) (mQ^4/M2^2 - mQ^2/M2 - 1),
   mixedSS/bigS^2 (-mQ^5/(4 M2^2) + mQ^3/(2 M2))}
};

rotation = {{Sin[theta], Cos[theta]}, {Cos[theta], -Sin[theta]}};
piBasis = {{piAA, piAB}, {piAB, piBB}};
piPhysical = Expand[rotation.piBasis.Transpose[rotation]];

(* Ward identity and leading tensor-current reduction. *)
wardDeterminant = Det[{{eta0, eta1, eta2, eta3}, {xi0, xi1, xi2, xi3},
  {q0, q1, q2, q3}, {q0, q1, q2, q3}}];
wardCheck = FullSimplify[wardDeterminant] === 0;
twist2TraceA = -8 mQ;
twist2TraceB = -8 k2/(mQ + ms);
twist2PreBorelRatio = Factor[twist2TraceB/twist2TraceA];
(* k2/(mQ^2-k2)=mQ^2/(mQ^2-k2)-1; the contact term has zero Borel image. *)
twist2PostBorelRatio = mQ/(mQ + ms);

formulaAssociation = <|
  "TraceAA" -> traceAA,
  "TraceAB" -> traceAB,
  "TraceBB" -> traceBB,
  "rhoAA" -> rhoAA,
  "rhoAB" -> rhoAB,
  "rhoBA" -> rhoBA,
  "rhoBB" -> rhoBB,
  "SpectralDeterminant" -> spectralDeterminant,
  "d3Matrix" -> d3Matrix,
  "d3msMatrix" -> d3msMatrix,
  "d5Matrix" -> d5Matrix,
  "PhysicalMatrix" -> piPhysical,
  "Twist2PreBorelRatio" -> twist2PreBorelRatio,
  "Twist2PostBorelRatio" -> twist2PostBorelRatio
|>;
Put[formulaAssociation, FileNameJoin[{outputDir, "analytic_formulas.wl"}]];

reportLines = {
  "Wolfram Kernel version: " <> ToString[$Version],
  "rhoAB-rhoBA == 0: " <> ToString[symmetryCheck],
  "Ward determinant == 0: " <> ToString[wardCheck],
  "Cut trace AA: " <> ToString[traceAA, InputForm],
  "Cut trace AB: " <> ToString[traceAB, InputForm],
  "Cut trace BB: " <> ToString[traceBB, InputForm],
  "rhoAA: " <> ToString[rhoAA, InputForm],
  "rhoAB: " <> ToString[rhoAB, InputForm],
  "rhoBB: " <> ToString[rhoBB, InputForm],
  "det(rho): " <> ToString[spectralDeterminant, InputForm],
  "d3 matrix: " <> ToString[d3Matrix, InputForm],
  "finite-ms d3 matrix: " <> ToString[d3msMatrix, InputForm],
  "d5 matrix: " <> ToString[d5Matrix, InputForm],
  "rotated matrix: " <> ToString[piPhysical, InputForm],
  "twist-2 pre-Borel B/A ratio: " <> ToString[twist2PreBorelRatio, InputForm],
  "twist-2 post-Borel B/A ratio: " <> ToString[twist2PostBorelRatio, InputForm]
};
Export[FileNameJoin[{outputDir, "symbolic_validation.txt"}], StringRiffle[reportLines, "\n"], "Text"];

nb = Notebook[{
  Cell["Mixed-current Ds1/Bs1 to Ds*/Bs* gamma derivation", "Title"],
  Cell["This notebook is generated by scripts/symbolic_derivation.wl. It contains no imported Python result.", "Text"],
  Cell["1. Conventions and currents", "Section"],
  Cell["Metric (+---), epsilon^0123=+1, p’=p+q, q^2=0. Basis currents: JA=bar(s) gamma_mu gamma5 Q and JB=i bar(s) sigma_mu_nu p’^nu gamma5 Q/(mQ+ms).", "Text"],
  Cell[BoxData@ToBoxes[{g0, g5}], "Input"],
  Cell["2. Cut Dirac traces", "Section"],
  Cell[BoxData@ToBoxes[{traceAA, traceAB, traceBB}], "Output"],
  Cell["3. Explicit spectral densities", "Section"],
  Cell[BoxData@ToBoxes[{rhoAA, rhoAB, rhoBB}], "Output"],
  Cell["The determinant is nonnegative for s >= (mQ+ms)^2.", "Text"],
  Cell[BoxData@ToBoxes[spectralDeterminant], "Output"],
  Cell["4. Local d=3 and d=5 Borel matrices", "Section"],
  Cell[BoxData@ToBoxes[d3Matrix + d3msMatrix + d5Matrix], "Output"],
  Cell["5. Physical rotation", "Section"],
  Cell[BoxData@ToBoxes[piPhysical], "Output"],
  Cell["6. External-photon transition", "Section"],
  Cell["The retained invariant multiplies i e epsilon(eta,xi*,epsilon*,q). Ordinary local transition condensates are excluded. Hard emission and BBK two-/three-particle DAs through twist four are retained.", "Text"],
  Cell["For the tensor current the explicit twist-two trace ratio is k^2/[mQ(mQ+ms)]. Rewriting k^2/(mQ^2-k^2)=mQ^2/(mQ^2-k^2)-1 and Borel-transforming removes the contact term, giving mQ/(mQ+ms). This leading-power projection is applied term by term at the declared truncation.", "Text"],
  Cell[BoxData@ToBoxes[{twist2PreBorelRatio, twist2PostBorelRatio}], "Output"],
  Cell["7. Ward identity", "Section"],
  Cell[BoxData@ToBoxes[wardDeterminant], "Output"],
  Cell["8. Numerical regression", "Section"],
  Cell["Run scripts/mathematica_regression.wl; its CSV is compared term by term with outputs/csv/python_regression.csv.", "Text"]
}];
```

```
Put[nb, FileNameJoin[{notebookDir, "Dsstar_Bsstar_gamma2_derivation.nb"}]];

Print[StringRiffle[reportLines, "\n"]];
Exit[0];
```

B Complete independent Mathematica regression source

This listing evaluates native numerical integrals and writes the Mathematica CSV. It never reads a Python result.

```
(* Independent Mathematica numerical implementation; no Python result is read. *)
ClearAll["Global‘*"];
Print["regression: start"];
scriptDir = DirectoryName[$InputFileName];
projectDir = DirectoryName[scriptDir];
csvDir = FileNameJoin[{projectDir, "outputs", "csv"}];
If[!DirectoryQ[csvDir], CreateDirectory[csvDir, CreateIntermediateDirectories -> True]];

alphaEM = 0.0072973525693;
esCharge = -1/3;
qq = -(0.24)^3;
kappaS = 0.8;
ss = kappaS qq;
m0sq = 0.8;
mixedSS = m0sq ss;
chi = -3.15;
f3 = -0.0039;
omegaV = 3.8;
omegaA = -2.1;
kappaDA = 0.2;
zeta1 = 0.4;
zeta2 = 0.3;
ms0 = 0.093;

(* Native adaptive integration at the declared regression points. *)
gIntegrate[f_, a_?NumericQ, b_?NumericQ] := NIntegrate[
  f[z], {z, a, b}, Method -> "GlobalAdaptive", AccuracyGoal -> 10,
  MaxRecursion -> 12
];

lambdaF[s_, m_, ms_] := (s - (m + ms)^2) (s - (m - ms)^2);
rhoAA[s_, m_, ms_] := Sqrt[lambdaF[s, m, ms]]/(8 Pi^2) *
  (s - (m + ms)^2) ((m - ms)^2 + 2 s)/s^2;
rhoAB[s_, m_, ms_] := Sqrt[lambdaF[s, m, ms]]/(8 Pi^2) *
  3 (m - ms) (s - (m + ms)^2)/((m + ms) s);
rhoBB[s_, m_, ms_] := Sqrt[lambdaF[s, m, ms]]/(8 Pi^2) *
  (s - (m + ms)^2) (s + 2 (m - ms)^2)/((m + ms)^2 s);
rhoV[s_, m_, ms_] := Sqrt[lambdaF[s, m, ms]]/(8 Pi^2) *
  (2 - (m^2 + ms^2 - 6 m ms)/s - (m^2 - ms^2)^2/s^2);

localMatrices[M2_, m_, ms_] := Module[{b = m + ms, e, d3, corr, v, d3ms, d5},
  e = Exp[-m^2/M2];
  d3 = ss e {{m, m^2/b}, {m^2/b, m^3/b^2}};
  corr = ss e (-m^2 ms/(2 M2) + m^3 ms^2/(2 M2^2));
  v = {1, m/b};
  d3ms = corr Outer[Times, v, v];
  d5 = e {
    {-mixedSS m^3/(4 M2^2), -mixedSS/(4 b) (m^4/M2^2 - m^2/M2 - 1)},
    {-mixedSS/(4 b) (m^4/M2^2 - m^2/M2 - 1),
     mixedSS/b^2 (-m^5/(4 M2^2) + m^3/(2 M2))}
  };
  {d3, d3ms, d5}
];

borelBasis[M2_, s0_, m_, ms_] := Module[{sth, p, locals},
  sth = (m + ms)^2;
  p = {
    {gIntegrate[(Exp[-#/M2] rhoAA[#, m, ms])&, sth, s0],
     gIntegrate[(Exp[-#/M2] rhoAB[#, m, ms])&, sth, s0]},
    {gIntegrate[(Exp[-#/M2] rhoAB[#, m, ms])&, sth, s0],
     gIntegrate[(Exp[-#/M2] rhoBB[#, m, ms])&, sth, s0]}
  };
  locals = localMatrices[M2, m, ms];
  p + Total[locals]
];
rot[thetaDeg_] := {{Sin[thetaDeg Degree], Cos[thetaDeg Degree]},
  {Cos[thetaDeg Degree], -Sin[thetaDeg Degree]}};
projected[M2_, s0_, m_, ms_, theta_] := rot[theta].borelBasis[M2, s0, m, ms].Transpose[rot[theta]];

fullProjected[M2_, m_, ms_, theta_] := Module[{sth, hi, p, locals},
  sth = (m + ms)^2; hi = sth + 60 M2;
  p = {
    {gIntegrate[(Exp[-#/M2] rhoAA[#, m, ms])&, sth, hi],
     gIntegrate[(Exp[-#/M2] rhoAB[#, m, ms])&, sth, hi]},
    {gIntegrate[(Exp[-#/M2] rhoAB[#, m, ms])&, sth, hi],
     gIntegrate[(Exp[-#/M2] rhoBB[#, m, ms])&, sth, hi]}
  };
  locals = Total[localMatrices[M2, m, ms]];
  rot[theta].p.Transpose[rot[theta]] + rot[theta].locals.Transpose[rot[theta]]
```

```
];
massSR[M2_, s0_, m_, ms_, theta_, state_] := Module[{tau, h, fp, fm, f0, r1},
  tau = 1/M2; h = Max[10^-5, tau 2 10^-4];
  fp = projected[1/(tau + h), s0, m, ms, theta][[state, state]];
  fm = projected[1/(tau - h), s0, m, ms, theta][[state, state]];
  f0 = projected[M2, s0, m, ms, theta][[state, state]];
  r1 = -(fp - fm)/(2 h);
  Sqrt[Max[r1/f0, 0]]
];
fInitial[M2_, s0_, m_, ms_, theta_, state_, mass_] :=
  Exp[mass^2/(2 M2)] Sqrt[projected[M2, s0, m, ms, theta][[state, state]]]/mass;

borelVector[M2_, s0_, m_, ms_] := Module[{sth, pert, locs, local},
  sth = (m + ms)^2;
  pert = gIntegrate[(Exp[-#/M2] rhoV[#, m, ms])&, sth, s0];
  locs = localMatrices[M2, m, ms];
  local = -Total[locs][[1, 1]];
  {pert, local, pert + local}
];
fVector[M2_, s0_, m_, ms_, mass_] :=
  Exp[mass^2/(2 M2)] Sqrt[borelVector[M2, s0, m, ms][[3]]]/mass;

phi[u_] := 6 u (1 - u);
hGamma[u_] := -10 (3 (2 u - 1)^2 - 1)/2;
psiV[u_] := 5 (3 (2 u - 1)^2 - 1) +
  3/64 (15 omegaV - 5 omegaA) (3 - 30 (2 u - 1)^2 + 35 (2 u - 1)^4);
psiA[u_] := (1 - (2 u - 1)^2) (5 (2 u - 1)^2 - 1) 5/2 *
  (1 + 9 omegaV/16 - 3 omegaA/16);
Atw4[u_] := 40 u^2 (1-u)^2 (3 kappaDA + 1) -
  24 zeta2 (u (1-u) (2 + 13 u (1-u)) +
   2 u^3 (10 - 15 u + 6 u^2) Log[u] +
   2 (1-u)^3 (10 - 15 (1-u) + 6 (1-u)^2) Log[1-u]);
HGamma[u_] := Integrate[hGamma[x], {x, 0, u}];
Hbar[u_] := Integrate[(u-x) hGamma[x], {x, 0, u}];
PsiV[u_] := Integrate[psiV[x], {x, 0, u}];
iF2 = 0;
iF3 = 15 (omegaA - 3 omegaV + 4)/32;

rhoT[s_, m_, ms_, eQ_] := Module[{root, aS, aQ},
  root = Sqrt[lambdaF[s, m, ms]];
  aS = s - m^2 + ms^2; aQ = s - ms^2 + m^2;
  3 ms m/(4 Pi^2) (esCharge Log[(aS-root)/(aS+root)] +
    eQ Log[(aQ-root)/(aQ+root)])
];
transitionTerms[M2_, s0_, m_, ms_, eQ_] := Module[
  {u0=1/2, sth, em, es0, diff, hard, local, tw2, tw42, tw32, tw43, tw33, total},
  sth=(m+ms)^2; em=Exp[-m^2/M2]; es0=Exp[-s0/M2]; diff=em-es0;
  hard=gIntegrate[(Exp[-#/M2] rhoT[#,m,ms,eQ])&,sth,s0];
  local=eQ m em ss (1-ms^2/M2 (1-m^2/M2));
  tw2=esCharge m ss diff M2 chi phi[u0];
  tw42=esCharge m ss em (-m^2 Atw4[u0]/(4 M2)-
    HGamma[u0](1-u0)-Hbar[u0](1-2 m^2/M2));
  tw32=esCharge f3 M2 diff ((1-u0) psiA'[u0]/4-psiA[u0]/4-
    PsiV[u0](1+2m^2/M2)+(1-u0)psiV[u0]);
  tw43=m esCharge ss em iF2;
  tw33=-esCharge f3 M2 diff iF3;
  total=hard+tw2+tw42+tw32+tw43+tw33;
  <|"hard"->hard,"tw2"->tw2,"tw4_2p"->tw42,"tw3_2p"->tw32,
    "tw4_3p"->tw43,"tw3_3p"->tw33,"local_heavy_diagnostic"->local,
    "quoted_total_A"->total,"quoted_total_B"->m/(m+ms) total|>
];

(* Reproduce the accepted-grid residue averages used by the publication path. *)
initialAverage[m_, theta_, state_, mass_, window_] := Module[
 {mgrid, sgrid, vals={}, M2, s0, pi0, pc, d5frac, msr, pieces, full},
 mgrid=Subdivide[window[[1]],window[[2]],5];
 sgrid=Subdivide[window[[3]],window[[4]],4];
 Do[
  pi0=projected[M2,s0,m,ms0,theta][[state,state]];
  full=fullProjected[M2,m,ms0,theta][[state,state]];
  pc=pi0/full;
  pieces=localMatrices[M2,m,ms0];
  d5frac=Abs[(rot[theta].pieces[[3]].Transpose[rot[theta]])[[state,state]]]/Abs[pi0];
  msr=massSR[M2,s0,m,ms0,theta,state];
  If[pi0>0 && pc>=0.40 && d5frac<=0.10 && Abs[msr/mass-1]<=0.05,
    AppendTo[vals,fInitial[M2,s0,m,ms0,theta,state,mass]]],
  {M2,mgrid},{s0,sgrid}];
 Mean[Flatten[vals]]
];
vectorAverage[m_, mass_, window_] := Module[{mgrid,sgrid,vals={},M2,s0,bv,sth,fullPert,full,pc,lf},
 mgrid=Subdivide[window[[1]],window[[2]],5];sgrid=Subdivide[window[[3]],window[[4]],4];
 Do[bv=borelVector[M2,s0,m,ms0];sth=(m+ms0)^2;
  fullPert=gIntegrate[(Exp[-#/M2]rhoV[#,m,ms0])&,sth,sth+60M2];
  full=fullPert+bv[[2]];pc=bv[[3]]/full;lf=Abs[bv[[2]]]/Abs[bv[[3]]];
  If[pc>=0.40&&lf<=0.25,AppendTo[vals,fVector[M2,s0,m,ms0,mass]]],
  {M2,mgrid},{s0,sgrid}];Mean[Flatten[vals]]
];

sectors = {
 <|"name"->"D","m"->1.27,"eQ"->2/3,"theta"->38.4,"mf"->2.1122,
   "vwindow"->{2.5,3.5,6.5,7.5},"test"->{3.0,8.0,5.0,6.75},
   "states"->{
```

```
    <|"key"->"Ds1_2460","index"->1,"mass"->2.4595,"tp"->{2.,3.,7.5,8.5}|>,
    <|"key"->"Ds1_2536","index"->2,"mass"->2.53512,"tp"->{2.,3.,8.5,10.}|>
   }|>,
 <|"name"->"B","m"->4.18,"eQ"->-1/3,"theta"->35.264,"mf"->5.4154,
   "vwindow"->{7.,9.,36.,38.},"test"->{9.0,42.0,6.0,40.0},
   "states"->{
    <|"key"->"Bs1_lower","index"->1,"mass"->5.75,"tp"->{8.,11.,40.,42.}|>,
    <|"key"->"Bs1_5830","index"->2,"mass"->5.82873,"tp"->{7.,9.,42.,44.}|>
   }|>
};
Print["regression: definitions complete"];

rows={{"sector","quantity","value"}};
Do[
 Print["regression: sector ", sec["name"]];
 name=sec["name"];m=sec["m"];eQ=sec["eQ"];theta=sec["theta"];mf=sec["mf"];
 {m2tp,s0tp,m2tr,s0tr}=sec["test"];
 sTest=(m+ms0)^2+1;
 rows=Join[rows,{{name,"rho_AA",N[rhoAA[sTest,m,ms0],16]},
   {name,"rho_AB",N[rhoAB[sTest,m,ms0],16]},
   {name,"rho_BB",N[rhoBB[sTest,m,ms0],16]}}];
 mat=borelBasis[m2tp,s0tp,m,ms0];
 Print["regression: basis matrix complete"];
 rows=Join[rows,{{name,"Pi_AA",mat[[1,1]]},{name,"Pi_AB",mat[[1,2]]},{name,"Pi_BB",mat[[2,2]]}}];
 terms=transitionTerms[m2tr,s0tr,m,ms0,eQ];
 Print["regression: transition terms complete"];
 Do[AppendTo[rows,{name,"T_"<>key,terms[key]}],
  {key,{"hard","tw2","tw4_2p","tw3_2p","tw4_3p","tw3_3p","quoted_total_A","quoted_total_B"}}];
 fv=fVector[Mean[sec["vwindow"][[1;;2]]],Mean[sec["vwindow"][[3;;4]]],m,ms0,mf];
 Print["regression: vector residue complete"];
 Do[
  fi=fInitial[m2tp,s0tp,m,ms0,theta,st["index"],st["mass"]];
  rr=rot[theta][[st["index"]]];
  tphys=rr[[1]]terms["quoted_total_A"]+rr[[2]]terms["quoted_total_B"];
  pref=Exp[(st["mass"]^2+mf^2)/(2m2tr)]/(fi fv st["mass"] mf);
  coupling=pref tphys;
  kg=(st["mass"]^2-mf^2)/(2st["mass"]);
  width=alphaEM coupling^2 kg^3 (st["mass"]^2+mf^2)/(3st["mass"]^2 mf^2) 10^6;
  rows=Join[rows,{{name,st["key"]<>"_f",fi},{name,st["key"]<>"_g",coupling},
    {name,st["key"]<>"_width_keV",width}}],
  {st,sec["states"]}];
 Print["regression: state residues and observables complete"];
 AppendTo[rows,{name,"f_"<>name<>"sstar",fv}],
 {sec,sectors}];

csvText = StringRiffle[
  (StringRiffle[(ToString[#, InputForm]& /@ #), ","]& /@ rows), "\n"];
Export[FileNameJoin[{csvDir,"mathematica_regression.csv"}],csvText,"Text"];
Print["regression: wrote ", Length[rows]-1, " rows"];
Exit[0];
```

C Reproduction commands

```
export PROJECT_PYTHON=/Users/sbilmis/.cache/codex-runtimes/codex-primary-runtime/dependencies/python/bin/python3
"$PROJECT_PYTHON" scripts/run_analysis.py
/Applications/Wolfram.app/Contents/MacOS/WolframKernel -noinit -noprompt -script scripts/symbolic_derivation.wl
/Applications/Wolfram.app/Contents/MacOS/WolframKernel -noinit -noprompt -script scripts/mathematica_regression.wl
"$PROJECT_PYTHON" scripts/compare_regression.py
"$PROJECT_PYTHON" scripts/build_documents.py
latexmk -pdf -interaction=nonstopmode -halt-on-error -cd tex/Dsstar_Bsstar_gamma2_paper.tex
latexmk -pdf -interaction=nonstopmode -halt-on-error -cd tex/Dsstar_Bsstar_gamma2_Mathematica_derivation.tex
```